

Temperature Humidity Luminance Controller — User Guide

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What It Does

The THL Controller monitors and automatically controls the environment inside a growing cabinet or room. It measures temperature, humidity, and CO₂, and uses that data to drive four devices:

Device	What it controls
Heater	Keeps temperature at your target
Humidifier	Keeps humidity at your target
Fan	Removes CO ₂ / provides timed ventilation
Light	Runs a daily on/off schedule with optional sunrise/sunset dimming

It logs all readings to an SD card, and can send instant alerts to your phone via the free **ntfy** app.

First-Time Setup

Step 1 — Connect to the controller's Wi-Fi

On first power-up (or after a factory reset), the controller creates its own Wi-Fi access point:

Setting	Value
Network name (SSID)	thlcontroller
Password	setup1234

Connect your phone or laptop to that network.

Step 2 — Open the settings page

Open a browser and go to **http://192.168.4.1**, then click **Settings / Logs** (the gear / log icon at the top of the page).

Step 3 — Enter your Wi-Fi credentials

Scroll down to the **WiFi & Device** card and fill in:

- **Primary SSID** — the name of your home/facility Wi-Fi network
- **Primary Password** — the password for that network
- **Fallback SSID / Password** (*optional*) — a second network to try if the first is unavailable (e.g. a mobile hotspot)

Click **Save WiFi & Device**. The controller will reboot and connect to your network.

Step 4 — Find the controller's new address

Once connected to your normal Wi-Fi, the controller is accessible at:

- **http://<hostname>.local** — e.g. `http://t-h-l-controller.local` (works on most phones and laptops)
- Or by its IP address, which you can find in your router's device list

Tip: Set a custom hostname in **WiFi & Device** settings so it is easy to remember, e.g. `growroom-1` → `http://growroom-1.local`

Connecting to the Interface

Open a browser and navigate to the controller's address. The interface has two main pages:

Page	How to get there
Dashboard (live view)	Click the home icon / device name at the top
Settings & Logs	Click the log/gear icon at the top

The interface updates the live readings automatically — no need to refresh.

Dashboard

The dashboard shows the current state of the environment and all devices.

Status bar (top right)

Indicator	Meaning
HH:MM (time)	Current device time (check timezone in Settings if wrong)
SD: OK (green)	SD card is mounted and logging
SD: ! (orange)	SD card problem — hover/tap for details
Door open (yellow)	Door switch is reporting the door as open
Sensors OK (green)	All sensors are reading successfully
⚠ Fault (red)	One or more sensors have stopped responding

Environment readings (top row)

Displays the current **Temperature (°C)**, **Humidity (%)**, **CO₂ (ppm)**, and **Water Level**. The water level card turns orange when the level is low.

Device cards

Four cards show the state of each controlled device:

- **ON / OFF** — whether the relay is currently switched
- **M** prefix (orange) — the device is in manual override mode (see [Hardware Test Page](#))
- Tap/click any card to open its settings

If a **TEST MODE ACTIVE** banner appears at the top, all automatic scheduling is suspended and relays are under manual control — remember to exit test mode when finished.

Device Controls

Tap any device card on the dashboard to open its settings modal. Changes take effect immediately and are saved to the device's memory.

Heater

Setting	Description
Temperature target	The temperature the heater aims to maintain (°C)
Hysteresis	Dead-band on either side of the target. The heater turns ON when temperature drops to <i>target - hysteresis</i> and turns OFF when it rises to <i>target + hysteresis</i> . A value of 1.0°C means the heater cycles between 23 °C and 25 °C when the target is 24 °C.

Safety: If temperature exceeds the configured maximum (default 32 °C), the heater is locked off regardless of the setpoint.

Humidifier

Setting	Description
Humidity target	The relative humidity (%) the humidifier maintains
Hysteresis	Dead-band around the target — works the same as the heater

Interlocks — the humidifier is automatically disabled when: - **Water level is low** — the water reservoir sensor has detected the tank is empty - **Fan is running** — to prevent humidity spiking during an extraction cycle

Fan / CO₂ Extraction

The fan has two independent methods of control — both can be active at the same time.

CO₂-triggered

Setting	Description
CO₂ threshold	Fan turns ON when CO ₂ reaches this level (ppm)
CO₂ hysteresis	Fan turns OFF when CO ₂ drops this many ppm below the threshold

Example: threshold 1200 ppm, hysteresis 100 ppm → fan ON at 1200, OFF at 1100.

Timed schedule

Useful for minimum fresh-air ventilation regardless of CO₂ level.

Setting	Description
Schedule enabled	Tick to activate timed ventilation
Fan ON duration	How long (minutes) the fan runs each cycle
Cycle period	Total time (minutes) between the start of each run

Example: ON duration 5 min, period 30 min → fan runs for 5 minutes every half-hour.

Light

Setting	Description
ON time	Time of day the light turns on (HH:MM, 24-hour)
OFF time	Time of day the light turns off (HH:MM, 24-hour)
Sunrise ramp	How long (minutes) the light takes to fade up to full brightness at turn-on. Set to 0 for instant-on.
Sunset ramp	How long (minutes) the light takes to fade down before turning off. Set to 0 for instant-off.
Colour temperature	Full — all LED rows active; Neutral — rows 1, 3, 5; Warm — rows 2, 4

The light will not run outside the ON–OFF window regardless of other settings.

Settings — WiFi & Device

Accessible from the Settings/Logs page.

Field	Description
Device Name	Display name shown at the top of the dashboard
Hostname	Network hostname. Set this to something memorable — the device will then be reachable as <code>http://<hostname>.local</code> on your network
Primary SSID / Password	Your main Wi-Fi network
Fallback SSID / Password	Optional second network tried if the primary is unavailable
Timezone	Select your region so timestamps in logs and alerts are correct

Saving these settings **reboots the device**. Wait ~15 seconds, then reconnect.

Wi-Fi reconnection behaviour: 1. Device tries the primary network 2. If unavailable after several attempts, tries the fallback network 3. If both fail, returns to the built-in access point (thlcontroller / setup1234, IP 192.168.4.1) so you can reconfigure without a factory reset

Settings — Push Notifications (ntfy)

The controller can send instant alerts to your phone using the free [ntfy](#) service — no account or sign-up required.

Setting up

1. **Install the ntfy app** on your phone:
 - Android: [Google Play](#)
 - iOS: [App Store](#)
2. **Choose a topic name** — this is like a private channel name. Pick something unique that only you know, such as growroom-doran-4821. It is case-sensitive.
3. **Subscribe to the topic** in the ntfy app (tap the + button and enter your topic name).
4. **Enter the same topic name** in the controller's Push Notifications card and click **Save**.
5. Tap **Send Test Notification** to confirm it is working — you should receive a message on your phone within a few seconds.

Privacy note: ntfy topics are not password-protected by default. Anyone who knows your topic name can send to it. Use a long, random-looking name to keep it private, or self-host the ntfy server.

Settings — Alert Conditions

Configure when the controller sends push notifications. All settings are saved to the device's non-volatile memory and survive reboots.

Alerts are only sent if a topic has been configured in Push Notifications.

Temperature

Sends an alert when temperature goes outside the low/high band.

Field	Description
Enabled	Turn this alert type on or off
Low threshold	Alert fires if temperature drops below this value (°C)
High threshold	Alert fires if temperature rises above this value (°C)
Repeat (min)	How often to re-send the alert while the condition persists

Humidity

Same structure as temperature, but for relative humidity (%).

CO₂

Same structure as temperature, but for CO₂ (ppm). Step size is 50 ppm.

Water Level

Sends an alert when the water reservoir is empty.

Field	Description
Enabled	Turn this alert type on or off
Repeat (min)	How often to re-send the alert while the level remains low

Door Open

Sends an alert if the door is left open for too long.

Field	Description
Enabled	Turn this alert on or off
Alert after (s)	Seconds the door must be open before the first alert fires
Repeat (s)	How often to re-send while the door remains open

SD Card Failure

Sends an alert when the SD card cannot be mounted or becomes unreadable.

Field	Description
Enabled	Turn this alert on or off
Repeat (min)	How often to re-send while the card remains inaccessible

WiFi Failure

Sends an alert when the device loses its network connection.

This alert can only be delivered once the device reconnects to the network. It is useful for detecting brief dropouts or for knowing when the device has recovered.

Field	Description
Enabled	Turn this alert on or off
Repeat (min)	How often to re-send while disconnected

Equipment Failure

Sends an alert when a device has been running for a sustained period but the environment has not improved — indicating a possible equipment fault.

Heater (can't reach temperature setpoint)

Fires if the **heater relay has been ON** but temperature is still below the setpoint after the configured time. The timer pauses during any fan cycle (when humidity and heating are interlocked) and only resets when the temperature actually reaches the setpoint.

Field	Description
Enabled	Turn this alert on or off
Alert after (min)	Minutes the heater must be running below setpoint before alerting
Repeat (min)	How often to re-send while the fault persists

Humidifier (can't reach humidity setpoint)

Fires if the **humidifier relay has been ON** but humidity is still below the setpoint after the configured time. The timer holds during fan cycles (when the humidifier is interlocked off) and only resets when humidity reaches the setpoint.

Field	Description
Enabled	Turn this alert on or off
Alert after (min)	Minutes the humidifier must be running below setpoint before alerting
Repeat (min)	How often to re-send while the fault persists

Fan / CO₂ not clearing

Fires if **CO₂ has remained above the threshold** for longer than the configured time — suggesting the extraction fan may have failed.

Field	Description
Enabled	Turn this alert on or off
Alert after (min)	Minutes CO ₂ must stay above threshold before alerting
Repeat (min)	How often to re-send while CO ₂ remains elevated

Power Fault / Reboot

Sends a single notification when the controller starts up (including after a power cut).

Field	Description
Enabled	Send a notification on every boot/restart

Settings — Cabinet Fan

The controller has a small internal fan to cool its own electronics. This is separate from the main extraction fan.

Field	Description
Fan ON above (°C)	Cabinet internal temperature at which the fan turns on
Hysteresis (°C)	The fan turns off when the temperature falls this many degrees below the threshold

The current panel temperature is shown on this page in real time. Default: ON above 40 °C, OFF below 35 °C.

Data Logs

The Logs page shows:

Recent Readings table

Displays the last ~1 hour of sensor and relay data (one row every 10 seconds) stored in device memory. Columns:

Column	Description
Time	Timestamp (ISO format once synced to NTP, or BOOT+Xs before sync)
°C	Temperature
%RH	Relative humidity
CO ₂	CO ₂ in ppm
Heat / Hum / Fan / Light	● = relay ON, ○ = relay OFF
Panel°C	Cabinet internal temperature

Click **Refresh** to load the latest data.

Log files (SD card)

Lists all CSV files stored on the SD card, one per day, in the format YYYY-MM-DD.csv. Click any file link to download it. Files can be opened in Excel or any spreadsheet application.

If the device had not yet synced to NTP when it started logging, data from around startup will be in a file called boot_log.csv.

Remount SD — if you removed the SD card briefly and reinserted it, use this button to make the controller re-read it without a full reboot.

Hardware Test Page

Navigate to **http://<device-address>/test** (not linked from the main UI — intended for commissioning and fault-finding only).

On this page you can:

- **Enable Test Mode** — suspends all automatic control
- **Force any relay ON or OFF** individually
- **Adjust light brightness** with a slider and preset buttons
- **Change light colour temperature** (Full / Neutral / Warm)
- **View the cabinet temperature** in real time

⚠ Always **disable Test Mode** when you have finished testing. While Test Mode is active, the heater, fan, and humidifier will not respond to sensor readings.

Factory Reset

Found at the bottom of the Settings/Logs page (red border card).

A factory reset **erases all stored settings**, including: - Wi-Fi credentials - All setpoints and schedules - ntfy topic name - All alert configuration

After the reset, the device reboots into access-point mode. Reconnect to thlcontroller (password setup1234) and visit <http://192.168.4.1> to reconfigure.

Only use factory reset as a last resort. If you have only lost Wi-Fi access, the device will automatically fall back to its own access point — try that first.

Troubleshooting

Can't reach the interface

- Check that your device is on the same Wi-Fi network as the controller
- Try <http://t-h-l-controller.local> (or whatever hostname you set)
- If mDNS doesn't work (common on some Android/Windows setups), find the IP address from your router
- If the device has lost Wi-Fi, connect to the thlcontroller network (password setup1234) and visit 192.168.4.1

Time shown is wrong

Open Settings → WiFi & Device and set the correct **Timezone**. The time is synced from the internet (NTP) automatically once connected to Wi-Fi — it does not use a battery-backed clock.

Logs show BOOT+Xs timestamps

The device had not yet synced its clock when those entries were recorded. This is normal at startup; once NTP sync completes the timestamps switch to normal date/time format automatically.

SD card not logging

- Check the **SD: !** indicator on the dashboard for error details
- Ensure the card is fully seated
- Use the **Remount SD** button on the Logs page before reaching for the card
- The controller keeps the last ~1 hour of readings in memory regardless, so short card absences do not cause data loss
- FAT32 format is required (most cards ship this way; maximum tested size is 32 GB)

No push notifications arriving

1. Confirm the topic name in Push Notifications settings matches exactly what you subscribed to in the ntfy app (case-sensitive)
2. Use **Send Test Notification** — if it returns an error code, check the device's Wi-Fi connection
3. Ensure the ntfy app has notification permission on your phone

Heater / humidifier running but environment not improving

Check the **Equipment Failure** alerts are enabled in Alert Conditions — the controller will notify you if a device appears to be running without effect. Also verify the **water level** card — the humidifier is locked off when the tank is empty.

Humidifier turns off during a fan cycle

This is by design. The fan interlock prevents the humidifier from running while extraction is active, to avoid false over-humidity readings and condensation. The humidifier resumes automatically when the fan stops.